

Drawing Block Diagrams and Factor Graphs with LaTeX and TikZ

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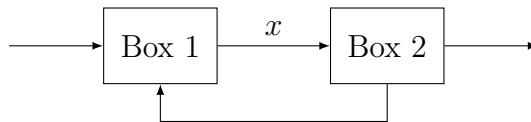
TikZ is a powerful package for drawing with LaTeX. This document provides some suggestions for a quick start with TikZ for drawing block diagrams and factor graphs.

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1 Block Diagrams

1.1 Getting Started



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\node[rectangle, draw, minimum width=15mm, minimum height=10mm]
  (box1) at (0,0) { Box~1 };
\node[rectangle, draw, minimum width=15mm, minimum height=10mm]
  (box2) at (30,0) { Box~2 };
\draw[->] (box1)+(-20,0) -- (box1);
\draw[->] (box1) -- node[above] {$x$} (box2);
\draw[->] (box2) -- +(20,0);
\draw[->] (box2) -- +(0,-10) -- +(-30,-10) -- (box1);
\end{tikzpicture}
```

Remarks:

- All examples require `\usepackage{tikz}` in the preamble.
- Arguments of `\begin{tikzpicture}`:
 - `x=1mm,y=1mm` changes the unit length from 1cm (default) to 1mm.
 - `>=latex` changes the default arrows to plain- $\text{\LaTeX}$ arrows.
- The two `\node` commands define nodes with names `box1` and `box2`, respectively. The coordinates refer to the center of the nodes by default.
- The second `\node` command can equivalently be written as

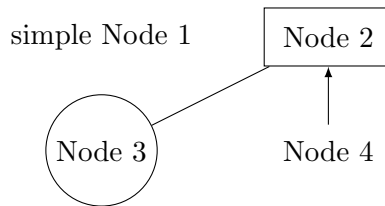
```
\draw (box1)+(30,0)
  node[rectangle, draw, minimum width=15mm, minimum height=10mm]
  (box2) { Box~2 };
```
- The last `\draw` command can equivalently be written in many different ways, e.g.:
 - with reversed arrows:

```
\draw[<-] (box1) -- +(0,-10) -- +(30,-10) -- (box2);
```
 - with step-by-step relative coordinates:

```
\draw[->] (box2) -- ++(0,-10) -- ++(-30,0) -- (box1);
```
 - with automatic corner positioning:

```
\draw[->] (box2) -- +(0,-10) -| (box1);
```

1.2 Positioning and Sizing of Nodes



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\draw (0,0) node (node1) { simple Node~1 };
\draw (node1)+(30,0) node[draw, inner sep=2.5mm] (node2) { Node~2 };
\draw (node1)+(0,-15) node[circle,draw,inner sep=1mm] (node3) { Node~3 };
\draw (node3|node2) node[inner sep=2mm] (node4) { Node~4 };
\draw (node3) -- (node2);
\draw[->] (node4) -- (node2);
\end{tikzpicture}
```

Remarks:

- The first `\draw` command can equivalently be written as
`\node (node1) at (0,0) { simple Node~1 };`
or simply as `\node (node1) { simple Node~1 };`
- The coordinates refer to the centers of the nodes by default.
- The position of `node4` can equivalently be specified as `(node2|-node3)`.

Positioning Relative to Borders:



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\node[rectangle, draw, minimum width=15mm, minimum height=10mm]
(box1) { Box~1 };
\node[rectangle, draw, minimum width=30mm, minimum height=10mm, right=20mm of box1]
(box2) { Box~2 };
\draw[<->] (box1) -- node[above]{20mm} (box2);
\draw[<->] (box2.east) -- node[above]{20mm} +(20,0);
\end{tikzpicture}
```

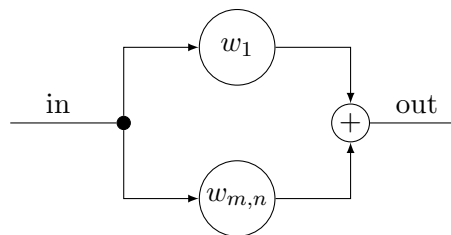
Remark: The second `\node` command requires `\usetikzlibrary{positioning}`.

1.3 Predefining Your Standard Nodes



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, minimum width=15mm, minimum height=10mm},
}
\draw (0,0) node[stdbox] (box1) { Box~1 };
\draw (box1)+(30,0) node[stdbox] (box2) { Box~2 };
\draw[->] (box1) -- (box2);
\end{tikzpicture}
```

More Interesting:



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  opcirc/.style = {draw, circle, inner sep = 0, minimum size = 5mm},
  stdcirc/.style = {draw, circle, minimum size = 10mm},
  blobcirc/.style = {draw, fill=black, circle, inner sep = 0, minimum size=1.75mm},
}
\draw (0,0) node[blobcirc] (branching) {};
\draw (branching)+(-15,0) -- (branching) node[pos=0.45, above] {in};
%
\draw (branching)+(15,10) node[stdcirc] (w1) {$w_1$};
\draw[->] (branching) |- (w1);
\draw (branching)+(15,-10) node[stdcirc] (w2) {\makebox(0,0){$w_{m,n}$}};
\draw[->] (branching) |- (w2);
%
\draw (branching)+(30,0) node[opcirc] (adder) {$+$};
\draw[->] (w1) -| (adder);
\draw[->] (w2) -| (adder);
\draw[->] (adder) -- +(15,0) node[pos=0.5, above]{out};
\end{tikzpicture}
```

Remark: The command `\makebox(0,0){$w_{m,n}$}` prevents the circle around $w_{m,n}$ to get too large. That whole line could alternatively be written as

`\draw (branching)+(15,-10) node[stdcirc, label={center:$w_{m,n}$}] (w2) {};`

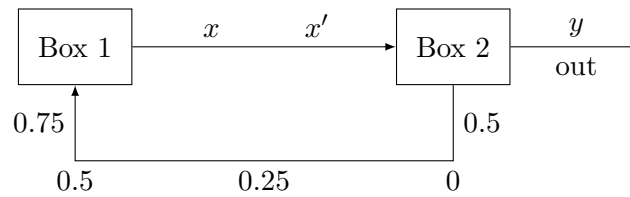
Nodes with Subpictures:



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  pics/crossbox/.style args={#1}{
    code={
      \node[rectangle, draw, thick,
        inner sep=0mm, minimum width=7.5mm, minimum height=7.5mm]
        (#1) at (0,0) {};
      \draw (#1.south west) -- (#1.north east);
      \draw[very thick, dashed] (#1.north west) -- (#1.south east);
    }
  },
}
\draw (0,0) pic{crossbox={Crossbox1}};
\draw (Crossbox1)+(25,0) pic{crossbox={Crossbox2}};
\draw[->] (Crossbox1) -- (Crossbox2);
\end{tikzpicture}
```

Remark: See also Section 1.6.

1.4 Positioning Along Paths

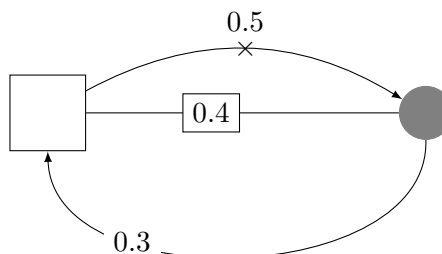


```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, minimum width=15mm, minimum height=10mm},
}
\draw (0,0) node[stdbox] (box1) { Box~1 };
\draw (box1)+(50,0) node[stdbox] (box2) { Box~2 };
\draw[->] (box1) -- node[above,pos=0.3]{$x$} node[above,pos=0.7] {$x'$} (box2);
\draw[->] (box2) -- node[above]{$y$} node[below]{out} +(25,0);
\draw[->] (box2.south) -- node[right,pos=0.5]{0.5} +(0,-10)
  -| (box1) node[below,pos=0]{0} node[below,pos=0.25]{0.25}
    node[below,pos=0.5]{0.5} node[left,pos=0.75]{0.75};
\end{tikzpicture}
```

Remarks:

- `pos` refers to the fractional position along the path.
- The last `\draw` command comprises two subpaths, each with `pos` ranging from 0 to 1. The second of these subpaths consists of two segments: in the first segment, `pos` ranges from 0 to 0.5; in the second segment, `pos` ranges from 0.5 to 1.

Placement On/Off the Lines:



```

\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, minimum width=10mm, minimum height=10mm},
  stddisc/.style = {draw, circle, fill, minimum size=7mm},
}
\draw (0,0) node[stdbox] (box) {};
\draw (50,0) node[stddisc, color=gray] (disc) {};
\draw[->] (box) to [bend left=30] node[above=1] { 0.5 } (disc);
\draw (box) -- node[draw, pos=0.4, fill=white] { 0.4 } (disc);
\draw[<-] (box) to [bend right=90] node[pos=0.3, fill=white] { 0.3 } (disc);
\end{tikzpicture}

```

Remark: Note the manual adjustment of the position of “0.5” above “ $\times$ ”.

Paths without the Lines:

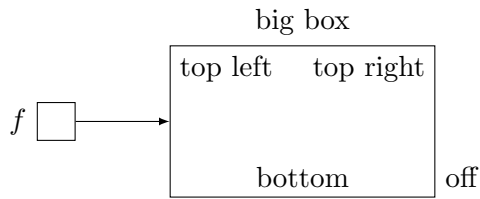


```

\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, minimum width=10mm, minimum height=10mm},
  stdcircle/.style = {draw, circle, minimum size=7mm},
  blobcirc/.style = {draw, fill=black, circle, inner sep = 0, minimum size=1.75mm},
}
\draw (0,0) node[stdbox] (box) {};
\draw (box)+(40,0) node[stdcircle] (circle) {};
\path (box) -- node[blobcirc,pos=0.33]{} node[blobcirc,pos=0.66]{} (circle);
\end{tikzpicture}

```

1.5 Labels and Positioning Around Nodes



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, minimum width=5mm, minimum height=5mm},
}
\draw (0,0) node[draw, rectangle, minimum width=35mm, minimum height=20mm,
  label={above:{big box}}] (Bigbox) {};
\draw (Bigbox.west)+(-15,0) node[stdbox,label={left:$f$}] (f) {};
\draw[->] (f) -- (Bigbox);
\draw (Bigbox.north west) node[anchor=north west] {top left};
\draw (Bigbox.north east) node[anchor=north east] {top right};
\draw (Bigbox.south) node[above] {bottom};
\draw (Bigbox.south east) node[above right] {off};
\end{tikzpicture}
```

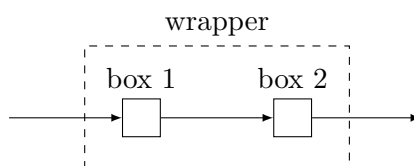
Adjusting Label Positions



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\node[draw, rectangle, minimum width=40mm, minimum height=15mm] (Bigbox) {};
\draw (Bigbox.north west) node[anchor=north west, xshift=2mm] {right};
\draw (Bigbox.south west) node[above left, yshift=2mm] {up};
\draw[->] (Bigbox.east) -- node[above=1mm, xshift=2mm]{hmm} +(30,0);
\end{tikzpicture}
```

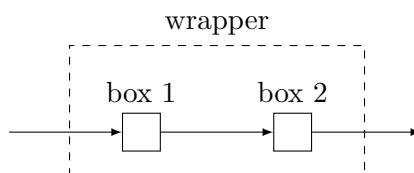
Remark: Use such adjustments only when it is clearly necessary. The default positions look good in most cases.

Box Around Stuff as a Node, Manual Sizing:



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, minimum width=5mm, minimum height=5mm},
}
\draw (0,0) node[stdbox,label={above:box~1}] (box1) {};
\draw (box1)+(20,0) node[stdbox,label={above:box~2}] (box2) {};
\draw[->] (box1)+(-17.5,0) -- (box1);
\draw[->] (box1) -- (box2);
\draw[->] (box2) -- +(17,0);
\draw (box1.south west)+(-5,-4) node[rectangle, draw, dashed, anchor=south west,
  minimum width=35mm, minimum height=16mm, label={above:wrapper}] {};
\end{tikzpicture}
```

Box Around Stuff as a Node, Automatic Sizing:

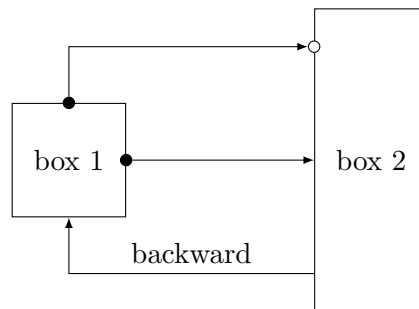


```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, minimum width=5mm, minimum height=5mm},
}
\draw (0,0) node[stdbox,label={[name=_box1] above:box~1}] (box1) {};
\draw (box1)+(20,0) node[stdbox,label={[name=_box2] above:box~2}] (box2) {};
\draw[->] (box1)+(-17.5,0) -- (box1);
\draw[->] (box1) -- (box2);
\draw[->] (box2) -- +(17,0);
\node[draw, dashed, inner sep=3.5mm,
  fit=(box1) (_box1) (box2) (_box2), label={above:wrapper}] {};
\end{tikzpicture}
```

Remarks:

- Automatic sizing requires `\usetikzlibrary{fit}` in the preamble.
- The labels `box1` and `box2` are themselves nodes, with labels `_box1` and `_box2`, respectively.
- Manual sizing may actually be easier to adjust properly.

Off-Center Connections:

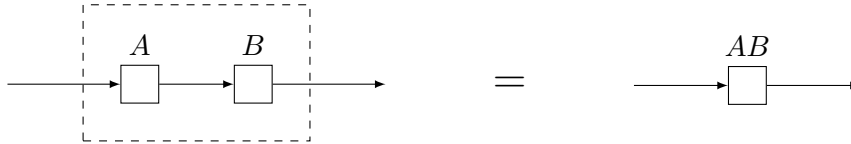


```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  sbdot/.style={draw, circle, minimum size=1.5mm, fill=black, inner sep=0mm},
  swdot/.style={draw, circle, minimum size=1.5mm, fill=white, inner sep=0mm},
}
\draw (0,0) node[rectangle, draw, minimum width=15mm, minimum height=15mm]
(box1) { box~1 };
\draw (40,0) node[rectangle, draw, minimum width=15mm, minimum height=40mm]
(box2) { box~2 };
%
\draw (box1.east) node[sbdot] (box1east) {};
\draw[->] (box1east) -- (box2.west|-box1east);
%
\draw (box1.north) node[sbdot] (box1north) {};
\draw (box2.west)+(0,15) node[swdot] (box2upwest) {};
\draw[->] (box1north) |- (box2upwest);
%
\draw (box2.west)+(0,-15) node[coordinate] (box2downwest) {};
\draw[->] (box2downwest) -| node[above, pos=0.25]{backward} (box1.south);
\end{tikzpicture}
```

Remark: `node[coordinate]` is an invisible empty node. That line can alternatively be written as `\coordinate[below=15mm of box2.west] (box2downwest);`

1.6 Notes with Subpictures or Plots

Subpictures as Nodes:



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  stdbox/.style = {draw, rectangle, anchor=center, inner sep=0mm,
    minimum width=5mm, minimum height=5mm},
}

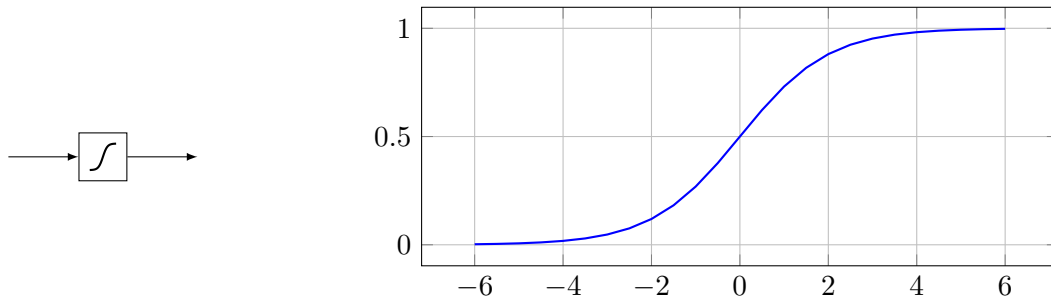
\node (leftpicture) {
  \begin{tikzpicture}
    \node[stdbox,label={above:$A$}](box1) {};
    \draw (box1)+(15,0) node[stdbox,label={above:$B$}](box2) {};
    \draw[->] (box1) -- (box2);
    \draw[<-] (box1) -- +(-17.5,0);
    \draw[->] (box2) -- +(17.5,0);
    %
    \draw (box1.south west)+(-5,-5) node[draw, rectangle, dashed,
      anchor=south west, minimum width=30mm, minimum height=18mm] {};
  \end{tikzpicture}
};

\draw (leftpicture.east)+(30,1)
  node[anchor=west] (rightpicture) {
    \begin{tikzpicture}
      \node[stdbox, label={above:$AB$}] (box) {};
      \draw[<-] (box) -- +(-15,0);
      \draw[->] (box) -- +(15,0);
    \end{tikzpicture}
  };

\path (leftpicture.east)
  -- node[pos=0.5, yshift=-2mm]{\Large$=$} (rightpicture.west);

\end{tikzpicture}
```

Plots Inside Nodes:



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]

\node[rectangle, draw] (simpleplot) {
  \begin{tikzpicture}[scale=1.0]
    \begin{axis}[hide axis, height=20mm, width=20mm, xscale=1.0, yscale=1.0]
      \addplot[domain=-6:6, thick] { 1/(1+exp(-x)) };
    \end{axis}
  \end{tikzpicture}
};
\draw[<-] (simpleplot) -- +(-12.5,0);
\draw[->] (simpleplot) -- +(12.5,0);

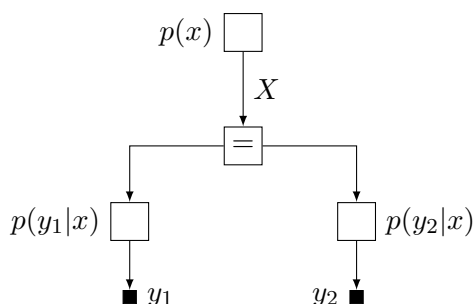
\draw (simpleplot.east)+(30,0)
  node[rectangle, anchor=west] (plotwithgrid) {
    \begin{tikzpicture}
      \begin{axis}[ grid=major, ytick={0,0.5,1}, height=5cm, width=10cm]
        \addplot[domain=-6:6, color=blue, thick] { 1/(1+exp(-x)) };
      \end{axis}
    \end{tikzpicture}
  };

\end{tikzpicture}
```

Remark: Requires `\usetikzlibrary{pgfplots}` and `\pgfplotsset{compat=1.17}` in the preamble. See the documentation of pgfplots to adjust the value of `compat`, if necessary.

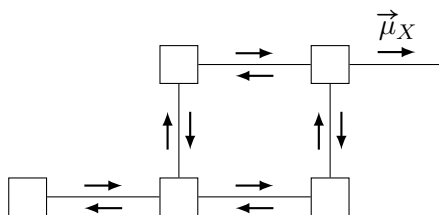
2 Factor Graphs

2.1 Example 1



```
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{
  opbox/.style = {draw, rectangle, inner sep=0mm,
    minimum width=5mm, minimum height=5mm},
  blobbox/.style = {draw, fill=black, rectangle, inner sep=0mm,
    minimum width=1.75mm, minimum height=1.75mm},
}
\draw (0,0) node[opbox] (equ) {$=$};
\draw (equ)+(0,15) node[opbox, label={left: $p(x)$}] (px) {};
\draw[->] (px) -- node[right]{$X$} (equ);
%
\draw (equ)+(-15,-10) node[opbox, label={left: $p(y_1 | x)$}] (py1cx) {};
\draw[->] (equ) -| (py1cx);
\draw (py1cx)+(0,-10) node[blobbox, label={right: $y_1$}] (y1) {};
\draw[->] (py1cx) -- (y1);
%
\draw (equ)+(15,-10) node[opbox, label={right: $p(y_2 | x)$}] (py2cx) {};
\draw[->] (equ) -| (py2cx);
\draw (py2cx)+(0,-10) node[blobbox, label={left: $y_2$}] (y2) {};
\draw[->] (py2cx) -- (y2);
\end{tikzpicture}
```

2.2 Example 2



```

\newcommand{\rightmsgarrow}{%
\begin{tikzpicture}
\draw[>, thick] (-2.5,0) -- (2.5,0);
\end{tikzpicture}}

\newcommand{\leftmsgarrow}{%
\begin{tikzpicture}
\draw[<-, thick] (-2.5,0) -- (2.5,0);
\end{tikzpicture}}

\newcommand{\upmsgarrow}{%
\begin{tikzpicture}
\draw[>, thick] (0,-2.5) -- (0,2.5);
\end{tikzpicture}}

\newcommand{\downmsgarrow}{%
\begin{tikzpicture}
\draw[<-, thick] (0,-2.5) -- (0,2.5);
\end{tikzpicture}}

\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\tikzset{ stdbox/.style = {draw, rectangle, inner sep=0mm,
                           minimum width=5mm, minimum height=5mm} }
\draw (0,0) node[stdbox] (box1) {};
\draw (20,0) node[stdbox] (box2) {};
\draw (40,0) node[stdbox] (box3) {};
\draw (20,17.5) node[stdbox] (box4) {};
\draw (40,17.5) node[stdbox] (box5) {};
\draw (box4) -- node[above] {\rightmsgarrow} node[below] {\leftmsgarrow} (box5);
\draw (box5) -- node[above=1.4mm,label={above:\msgf{\mu}{X}$},inner sep=0mm]
               {\rightmsgarrow} +(15,0);
\draw (box4) -- node[left] {\upmsgarrow} node[right] {\downmsgarrow} (box2);
\draw (box5) -- node[left] {\upmsgarrow} node[right] {\downmsgarrow} (box3);
\draw (box1) -- node[above] {\rightmsgarrow} node[below] {\leftmsgarrow} (box2);
\draw (box2) -- node[above] {\rightmsgarrow} node[below] {\leftmsgarrow} (box3);
\end{tikzpicture}

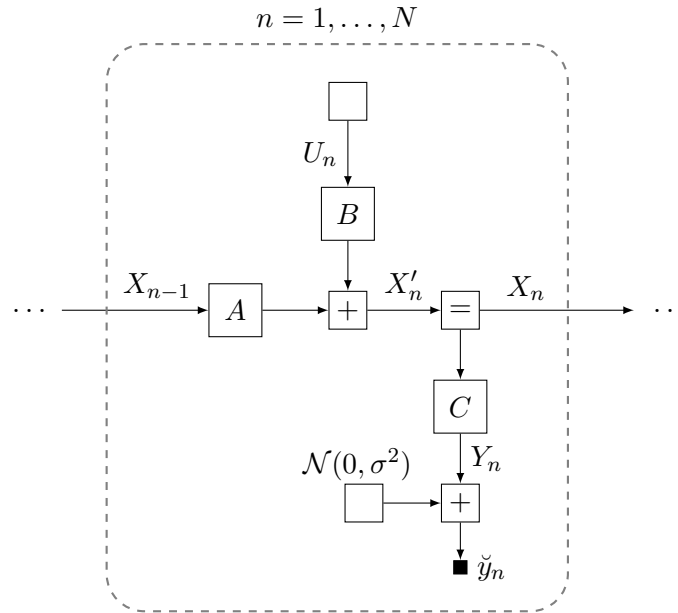
```

2.3 LaTeX Commands for Messages and Message Parameters

The commands `\msgf` and `\msgb` provided below allow to write $\vec{\mu}_X, \overleftarrow{\mu}_X, \vec{V}_Z, \overleftarrow{m}_Y$ as `\msgf{\mu}{X}`, `\msgb{\mu}{X}`, `\msgf{V}{Z}`, `\msgb{m}{Y}`\$.

```
% Definition of \msgf and \msgb for messages and message parameters
% by Christoph Reller
\makeatletter
\DeclareFontFamily{U}{MnSymbolA}{}
\DeclareSymbolFont{MnSyA}{U}{MnSymbolA}{m}{n}
\DeclareFontShape{U}{MnSymbolA}{m}{n}{
<-6> MnSymbolA5
<6-7> MnSymbolA6
<7-8> MnSymbolA7
<8-9> MnSymbolA8
<9-10> MnSymbolA9
<10-12> MnSymbolA10
<12-> MnSymbolA12}{}
\DeclareMathSymbol{\smallrightarrow}{\mathrel}{MnSyA}{0}
\DeclareMathSymbol{\smallleftarrow}{\mathrel}{MnSyA}{2}
\DeclareMathSymbol{\smallleftrightharpoonup}{\mathrel}{MnSyA}{16}
\newcommand{\smallrightarrowfill@}{\arrowfill@\relbar\relbar\smallrightarrow}
\newcommand{\smallleftarrowfill@}{\arrowfill@\smallleftarrow\relbar\relbar}
\newcommand{\smallleftrightharpoonupfill@}{
\arrowfill@\smallleftarrow\relbar\smallrightarrow}
\renewcommand{\overrightarrow}{\mathpalette{\overarrow@\smallrightarrowfill@}}
\renewcommand{\overleftarrow}{\mathpalette{\overarrow@\smallleftarrowfill@}}
\renewcommand{\overleftrightharpoonup}{
\mathpalette{\overarrow@\smallleftrightharpoonupfill@}}
\makeatother
\providecommand{\msgf}[2]{\protect\overrightarrow{#1}_\mspace{-3mu}#2}
\providecommand{\msgb}[2]{\protect\overleftarrow{#1}_\mspace{-3mu}#2}
```

2.4 Example 3



```

\begin{tikzpicture}[x=1mm, y=1mm, >=latex, scale=0.85]
\tikzset{
  opbox/.style = {draw, rectangle, inner sep=0mm,
    minimum width=5mm, minimum height=5mm},
  stdbox/.style = {draw, rectangle, minimum width=7mm, minimum height=7mm},
  blobbox/.style = {draw, fill=black, rectangle, inner sep=0mm,
    minimum width=1.75mm, minimum height=1.75mm},
}
\draw (0,0) node[stdbox] (A) {$A$};
\draw[->] (A)+(-27,0) -- (A) node[pos=0.65, above]{$X_{n-1}$};
\draw (A.west)+(-27.5,0) node { \ldots };
\draw (A)+(17.5,0) node[opbox] (plus) {$+$};
\draw[->] (A) -- (plus);
%
\draw (plus)+(0,15) node[stdbox] (B) {$B$};
\draw[->] (B) -- (plus);
\draw (B)+(0,17.5) node[opbox] (inpnoise) {};
\draw[->] (inpnoise) -- (B) node[pos=0.5, left] {$U_n$};
%
\draw (plus)+(17.5,0) node[opbox] (equ) {$=$};
\draw[->] (plus) -- (equ) node[pos=0.5, above] {$X'_n$};
\draw (equ)+(0,-15) node[stdbox] (C) {$C$};
\draw[->] (equ) -- (C);
\draw (C)+(0,-15) node[opbox] (outplus) {$+$};
\draw (outplus)+(-15,0) node[opbox] (obsnoise) {};

```

```

\draw[->] (obsnoise) -- (outplus);
\draw (obsnoise)+(-1,2) node[above] {$\mathcal{N}(0, \sigma^2)$};
\draw[->] (C) -- (outplus) node[pos=0.5, right] {$Y_n$};
\draw (outplus)+(0,-10) node[blobbox] (blob) {};
\draw[->] (outplus) -- (blob);
\draw (blob)+(1,0) node[right] {$\breve{y}_n$};
%
\draw[->] (equ)-- +(27,0) node[pos=0.3, above] {$X_n$};
\draw (equ.east)+(30,0) node { \ldots };

\draw (A.west |- blob) node (bottomleft) {}; % invisible position
\draw (bottomleft)+(-16,-7) node[draw, anchor=south west,
dashed, thick, color=gray, rounded corners=5mm,
minimum width=61mm, minimum height=75mm,
label={north:$n=1,\ldots,N$}] {};

\end{tikzpicture}

```

3 Moving On

TikZ can do very much more, for which the reader is referred to the original documentation and to other tutorials, cf. Section 3.3. We here conclude with two topics of a different nature.

3.1 Including Standalone Graphics

It often makes sense to create graphics as separate files using the `standalone` document class. That applies, in particular, to nontrivial plots, or plots of nontrivial data. Example:

```
\documentclass[tikz]{standalone}
\begin{document}
\begin{tikzpicture}[x=1mm, y=1mm, >=latex]
\draw (0,0) node[rectangle, draw, minimum width=15mm, minimum height=10mm] { };
\end{tikzpicture}
\end{document}
```

Suppose the file `mypicture.tex` contains the above. This file can be compiled independently, but it can also be included in the main LaTeX source with the command

```
\includestandalone[mode=buildnew]{mypicture},
```

which requires `\usepackage{standalone}` in the preamble.

3.2 Exporting SVG Graphics for HTML

A good way to embed TikZ pictures in HTML is via a .svg file. The .svg file can be produced as follows:

1. Put the source of your picture in a standalone .tex file using
`\documentclass[dvisvgm,tikz]{standalone}`; for example:

```
\documentclass[dvisvgm,tikz]{standalone} % use LaTeX, not pdflatex
\begin{document}
\begin{tikzpicture}[x=1mm, y=1mm, >=latex, thick]
\node[rectangle, draw, minimum width=15mm, minimum height=10mm]
  (box1) at (0,0) { Box~1 };
\node[rectangle, draw, minimum width=15mm, minimum height=10mm]
  (box2) at (30,0) { Box~2 };
\draw[->] (box1)+(-20,0) -- node[above]{$U$} (box1);
\draw[->] (box1) -- (box2);
\draw[->] (box2) -- node[above]{$Y$} +(20,0);
\draw[->] (box2) -- +(0,-10) -- +(-30,-10) -- (box1);
\end{tikzpicture}
\end{document}
```

2. Run LaTeX (not pdflatex) on the above to produce a .dvi file.
3. Run `dvisvgm --no-fonts` on the .dvi file.

For the above to work, you may need to update your LaTeX. (Old versions of LaTeX may complain.)

The resulting .svg file can be accessed from HTML, e.g., using `<img>`; for example:

```
<!DOCTYPE html> <html lang="en">
<head></head>
<body>
<header>
<h1>Displaying an SVG File</h1>
</header>

<center>

</center>

</body>
</html>
```

3.3 Links to Resources on LaTeX and TikZ

LaTeX:

- <https://www.latex-project.org/>
- <https://tobi.oetiker.ch/lshort/lshort.pdf>
- <https://moser-isi.ethz.ch/manuals.html#eqlatex>
- <https://latex-tutorial.com/color-latex/>

Easy Introductions to TikZ:

- <https://tikz.org/examples/chapter-02-creating-the-first-tikz-images/>
- <https://latexdraw.com/draw-a-rectangle-in-tikz/#t-1618533406677>
- <https://tikz.org/examples/chapter-03-drawing-positioning-and-aligning-nodes/>

TikZ and PGF Manual:

- <https://pgf-tikz.github.io/pgf/pgfmanual.pdf>
- <https://tikz.dev/>